EXPERIMENT-05

Aim: Create a content provider to manage and share data between different components of your android application.

To manage and share data between different components of your Android application, you can implement a ContentProvider. A ContentProvider acts as an abstraction layer over your app's data sources (like SQLite databases, files, or even network data) and provides a standard interface for other components or apps to query and modify that data.

Theory:
A Content Provider is an Android component used to manage and share data between applications. It provides a structured interface to access data using URIs and supports operations like insert, query, update, and delete.

1. Create Content Provider Class
   - Open Android Studio
   - Right-click package → New → Java Class → MyProvider
   - Extend ContentProvider class
   - Implement methods:
     - onCreate() → Initialize data (ArrayList with names)
     - query() → Return data using MatrixCursor
     - insert() → Add new data to list
     - delete() → Clear all data
     - update() and getType() → default return
2. Register Provider
   - Add in AndroidManifest.xml:

```
<provider android:name=".MyProvider" android:authorities="com.example.provider" android:exported="true"/>
```

3. Design Layout
   - In activity_main.xml, add:
     - EditText → Enter name
     - Button → Insert
     - Button → Show Data
4. Implement MainActivity
   - Initialize views using findViewById()
   - Define URI:

```
Uri uri = Uri.parse("content://com.example.provider");
```

   - Insert Button:
     - Use ContentValues
     - Call getContentResolver().insert()
   - Show Button:
     - Use query()
     - Read data using Cursor

- Display using Toast

5. Run Application
   - Enter name → Click Insert
   - Click Show Data → Display stored names

Code:

MyProvider.java

activity_main.xml | MainActivity.java | AndroidManifest.xml | MyProvider.java

```java
package com.example.experiment_05;

import android.content.ContentProvider;
import android.content.ContentValues;
import android.database.Cursor;
import android.database.MatrixCursor;
import android.net.Uri;
import java.util.ArrayList;

public class MyProvider extends ContentProvider {
    6 usages
    ArrayList<String> list = new ArrayList<>();
    @Override
    public boolean onCreate() {
        list.add("Ram");
        list.add("Shyam");
        list.add("Amit");
        return true;
    }
    @Override
    public Cursor query(Uri uri, String[] projection, String selection,
                        String[] selectionArgs, String sortOrder) {
        MatrixCursor cursor = new MatrixCursor(new String[]{"name"});
        for(String item : list){
            cursor.addRow(new Object[]{item});
        }
        return cursor;
    }
    @Override
    public Uri insert(Uri uri, ContentValues values) {
        String name = values.getAsString( key: "name");
        list.add(name);
        return uri;
    }
    @Override
    public int delete(Uri uri, String selection, String[] selectionArgs) {
        list.clear();
        return 1;
    }
    @Override
    public int update(Uri uri, ContentValues values, String selection,
                      String[] selectionArgs) {
        return 0;
    }
    @Override
    public String getType(Uri uri) {
        return null;
    }
}
```

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/editText"
        android:hint="Enter Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnInsert"
        android:text="Insert"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/btnShow"
        android:text="Show Data"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```java
package com.example.experiment_05;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.content.ContentValues;
import android.database.Cursor;
import android.net.Uri;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {
    EditText editText;
    Button btnInsert, btnShow;
    Uri uri = Uri.parse("content://com.example.provider");
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editText = findViewById(R.id.editText);
        btnInsert = findViewById(R.id.btnInsert);
        btnShow = findViewById(R.id.btnShow);
        btnInsert.setOnClickListener(v -> {
            ContentValues values = new ContentValues();
            values.put("name", editText.getText().toString());
            getContentResolver().insert(uri, values);
            Toast.makeText(this, "Data Inserted",Toast.LENGTH_SHORT).show();
        });
        btnShow.setOnClickListener(v -> {
            Cursor cursor = getContentResolver().query(uri, null, null, null, null);
            String result="";
            while(cursor.moveToNext()){
                result += cursor.getString(0) + "\n";
            }

            Toast.makeText(this,result,Toast.LENGTH_LONG).show();
        });
    }
}
```

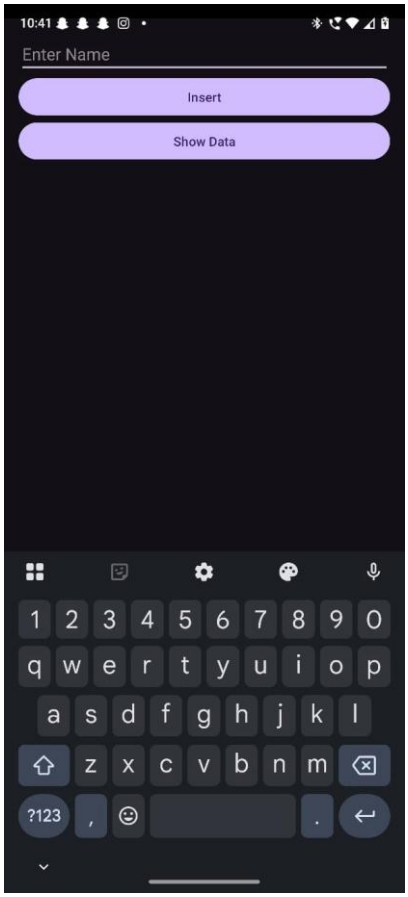
AndroidManifest.xml

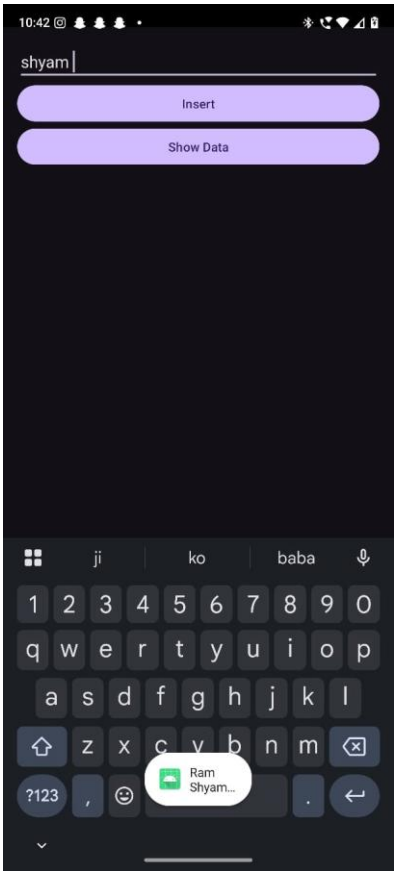
activity_main.xml | MainActivity.java | AndroidManifest.xml | MyProvider.java

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="Experiment-05"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Experiment05">

        <provider
            android:name=".MyProvider"
            android:authorities="com.example.provider"
            android:exported="true"/>

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Output:

1. Data Storage
   - Names are stored in an internal list inside Content Provider
2. Insert Operation
   - User enters name
   - Data is inserted using ContentValues
3. Data Retrieval
   - Data fetched using Cursor via query()
4. Display Output
   - All stored names shown using Toast message
5. Data Management
   - Supports:
   - Insert → add data
   - Query → fetch data
   - Delete → remove all data





Conclusion:

This practical demonstrates how a Content Provider manages and shares structured data using URIs and allows different components or apps to perform CRUD operations efficiently.